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Systems and methods for cooking pizza

Abstract

In general, cooking devices having at least one electric heating element and that are configured to cook various foods, including pizza, are provided. In some embodiments, the cooking device can include a housing having a base, a movable door coupled to the base that together define an interior cooking chamber. A cooking surface, such as a cooking stone, can be disposed proximate a heating element within the cooking chamber such that food, such as a pizza, can be placed on top of the cooking stone when inserted into the interior chamber. The heating element can be in operable communication with a controller configured to adjust the amount of heat supplied by the heating element to the interior chamber to optimize the cooking of a food, such as pizza, placed inside the interior chamber.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a divisional of U.S. patent application Ser. No. 18/307,595, filed Apr. 26, 2023, and entitled “SYSTEMS AND METHODS FOR COOKING PIZZA”, which is incorporated by reference herein in its entirety.

FIELD

(1) An oven system for cooking food, such as pizza, and methods for using the same, are provided.

BACKGROUND

(2) Pizzas can be made in a variety of styles (e.g., Neapolitan, New York Style, etc.) and can be prepared and cooked using techniques that vary based on the desired type of pizza style. Pizza ovens can provide for the cooking of pizza by using one or more heating elements to transfer heat to the pizza. However, some pizza ovens, such as those that use electric heating elements, may be limited in the ways by which they can transfer heat to the pizza during cooking and thus may be unable to cook a pizza in accordance with the cooking technique required to achieve certain pizza styles. For example, to successfully cook Neapolitan pizzas, the pizza oven must be capable of imparting a high level of heat to the crust of the pizza. Given the high levels of heat needed to cook such a pizza, it can be challenging to repeatedly cook Neapolitan pizzas with a short cycle time.

(3) Accordingly, there remains a need for an improved oven system for cooking various foods, including pizza.

SUMMARY

(4) An oven system for cooking food, such as pizza, and methods for using the same, are provided. Related apparatuses and techniques are also described.

(5) In one aspect, a cooking device is described. The cooking device can include a housing defining an interior chamber, an upper heating element disposed within the interior chamber and on an upper surface of the housing and configured to supply heat to the interior chamber. The cooking device can also include a lower heating element disposed within the interior chamber, on a lower surface of the housing, proximate a cooking surface disposed within the interior chamber and configured to receive food to be cooked, and the lower heating element can be configured to supply heat to the interior chamber. The lower surface can be opposite the upper surface. The cooking device can also include a controller in operable communication with the upper heating element and the lower heating element. The controller can be configured to receive temperature data characterizing a temperature of air within an upper region of the interior chamber above a cooking surface and a temperature of air within a lower region of the interior chamber below the cooking surface, to determine a ratio between an amount of heat to be supplied to the interior chamber by the upper heating element and an amount of heat to be supplied to the interior chamber by the lower heating element based on the temperature data, and to cause the heat to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined ratio.

(6) One or more of the following features can be included in any feasible combination. For example, the controller can be configured to receive data characterizing a mode of cooking the food and to determine the ratio based on the received data. For example, the cooking device can further

include an upper temperature sensor disposed within the interior chamber and proximate the upper heating element, and the upper temperature sensor can be configured to measure the temperature of the air within the interior chamber above the cooking surface. For example, the cooking device can further include a lower temperature sensor disposed within the interior chamber and proximate the lower heating element and the cooking surface, and the lower temperature sensor can be configured to measure the temperature of at least one of the cooking surface and the air within the lower region of the interior chamber. For example, the controller can be configured to cause the heat to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined ratio when the controller is operating in a pre-heating state. For example, the controller can be configured to adjust the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element after a period of time, and the period of time can be determined based on the determined ratio. For example, the controller can be configured to adjust the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element when the temperature of the air within the lower region of the interior chamber is less than a target temperature.

(7) In another aspect, a system is provided and can include at least one data processor and memory storing instructions configured to cause the at least one data processor to perform operations described herein. The operations can include receiving temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device, the upper region above a cooking surface, and characterizing a temperature of air within a lower region of the interior chamber, the lower region below the cooking surface; determining, based on the received temperature data, a ratio between an amount of heat to be supplied to the interior chamber by an upper heating element and an amount of heat to be supplied to the interior chamber by a lower heating element, the upper heating element and the lower heating element being disposed in the interior chamber at opposing surfaces of the interior chamber; and causing heat to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined ratio.

(8) One or more of the following features can be included in any feasible combination. For example, the operations can further include receiving mode data characterizing a mode of cooking the food, and wherein the ratio is determined based on the received mode data. For example, the causing of the heat to be supplied to the interior chamber can occur when the cooking device is operating in a pre-heating state. For example, the operations can further include determining a period of time based on the determined ratio, the period of time beginning after the causing of the heat to be supplied to the interior chamber; and adjusting the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element after the determined period of time has elapsed. For example, the operations can further include adjusting the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element when the temperature of the air within the lower region of the interior chamber characterized by the temperature data is less than a target temperature. For example, the operations can further include receiving second temperature data characterizing a second temperature of the air within the interior chamber; determining whether the second temperature characterized by the received second temperature data is greater than or equal to the target temperature; and, in response to determining that the temperature characterized by the received second temperature data is greater than or equal to the target temperature, causing the upper heating element and the lower heating element to deactivate. For example, the operations can further include determining a second period of time based on the determined ratio, the second period of time different from the period of time and beginning after the causing of the heat to be supplied to the interior chamber; determining, after the second period of time has elapsed, a second ratio

between the amount of heat to be supplied to the interior chamber by the upper heating element and the amount of heat to be supplied to the interior chamber by the lower heating element; and causing heat to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined second ratio.

(9) In another aspect, temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device above a cooking surface and configured to receive food to be cooked and a temperature of air within a lower region of the interior chamber below the cooking surface can be received. A ratio between an amount of heat to be supplied to the interior chamber by an upper heating element and an amount of heat to be supplied to the interior chamber by a lower heating element can be determined, and the upper heating element and the lower heating element can be disposed in the interior chamber at opposing surfaces of the interior chamber. Heat can be caused to be supplied to the interior chamber by at least one of the upper and lower heating elements based on the determined ratio.

(10) One or more of the following features can be included in any feasible combination. For example, mode data characterizing a mode of cooking the food can be received, and the ratio can be determined based on the received mode data. For example, the causing of the heat to be supplied to the interior chamber can occur when the cooking device is operating in a pre-heating state. For example, a period of time beginning after the causing of the heat to be supplied to the interior chamber can be determined based on the determined ratio, and the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element can be adjusted after the determined period of time has elapsed. For example, the amount of the heat supplied to the interior chamber by the upper heating element and the amount of the heat supplied to the interior chamber by the lower heating element can be adjusted when the temperature of the air within the lower region of the interior chamber characterized by the temperature data is less than a target temperature. For example, second temperature data characterizing a second temperature of the air within the upper region of the interior chamber can be received, whether the second temperature is greater than or equal to the target temperature can be determined, and, the upper heating element and the lower heating element can be deactivated in response to determining that the second temperature is greater than or equal to the target temperature.

(11) In one aspect, a cooking device is described. The cooking device can include a housing defining an interior chamber, an upper heating element disposed within the interior chamber and on an upper surface of the housing and configured to supply heat to the interior chamber. The cooking device can also include a lower heating element disposed within the interior chamber, on a lower surface of the housing, proximate a cooking surface disposed within the interior chamber and configured to receive food to be cooked, and the lower heating element can be configured to supply heat to the interior chamber. The lower surface can be opposite the upper surface. The cooking device can also include a controller in operable communication with the upper heating element and the lower heating element. The controller can be configured to cause a first amount of heat to be generated by the upper heating element and no heat to be generated by the lower heating element when the controller is operating in a first mode, and, when a temperature of the air within an upper region of the interior chamber exceeds a predetermined threshold and when the controller is operating in the first mode, cause the controller to operate in a second mode in which a second amount of heat is generated by the upper heating element and a third amount of heat is generated by the lower heating element, the second and third amounts of heat determined by the controller and based on the temperature of the air within the upper region of the interior chamber and a temperature of a lower region of the interior chamber located below the cooking surface.

(12) One or more of the following features can be included in any feasible combination. For example, the controller can be configured to receive data characterizing a mode of cooking the food and to determine the second and third amounts of heat based on the received data. For example, the

cooking device can further include an upper temperature sensor disposed within the interior chamber and proximate the upper heating element, and the upper temperature sensor can be configured to measure the temperature of the air within the upper region of the interior chamber. For example, the cooking device can further include a lower temperature sensor disposed within the interior chamber and proximate the lower heating element and the cooking surface, and the lower temperature sensor can be configured to measure the temperature of at least one of the lower region of the interior chamber and the cooking surface. For example, the controller can be configured to cause the heat to be supplied to the interior chamber by at least one of the upper and lower heating elements when the controller is operating in a pre-heating mode. For example, the controller can be configured to adjust the second and third amounts of heat after a period of time, and the period of time can be determined based on the temperature of the air within the upper region of the interior chamber and the temperature of the air within the lower region of the interior chamber. For example, the controller can be configured to adjust the second and third amounts of heat when the temperature of the cooking surface is less than a target temperature. For example, the controller can be configured to deactivate the upper heating element when the temperature of the air within the interior chamber exceeds the predetermined threshold.

(13) In another aspect, a system is provided and can include at least one data processor and memory storing instructions configured to cause the at least one data processor to perform operations described herein. The operations can include receiving temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device, the upper region above a cooking surface, and characterizing a temperature of air within a lower region of the interior chamber, the lower region below the cooking surface; determining, based on the received temperature data, a first amount of heat to be supplied to the interior chamber by an upper heating element and a second amount of heat to be supplied to the interior chamber by a lower heating element, the upper heating element and the lower heating element being disposed in the interior chamber at opposing surfaces of the interior chamber; determining, during an operating mode in which a third amount of heat is supplied to the interior chamber by the upper heating element and no heat is supplied to the interior chamber by the lower heating element, whether the temperature of the air within the upper region of the interior chamber exceeds a predetermined threshold; and in response to determining that the temperature of the air within the upper region of the interior chamber exceeds the predetermined threshold, causing the first and second amounts of heat to be supplied to the interior chamber.

(14) One or more of the following features can be included in any feasible combination. For example, the operations can further include receiving mode data characterizing a mode of cooking the food, and wherein the first and second amounts of heat are determined based on the received mode data. For example, the operations can further include adjusting the second and third amounts of heat after a period of time, and wherein the period of time is determined based on the temperature of the air within the upper region of the interior chamber and the temperature of the air within the lower region of the interior chamber. For example, the operations can further include adjusting the second and third amounts of heat when the temperature of the cooking surface is less than a target temperature. For example, the operations can further include causing the upper heating element and the lower heating element to deactivate in response to determining that the temperature of the air within the upper region of the interior chamber exceeds the predetermined threshold. For example, a portion of the temperature data can be received from a lower temperature sensor disposed within the interior chamber and proximate the lower heating element and the cooking surface, and the lower temperature sensor can be configured to measure the temperature of at least one of the lower region of the interior chamber and the cooking surface.

(15) In another aspect, temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device, the upper region above a cooking surface, and characterizing a temperature of air within a lower region of the interior chamber, the lower region

below the cooking surface can be received. A first amount of heat to be supplied to the interior chamber by an upper heating element and a second amount of heat to be supplied to the interior chamber by a lower heating element, the upper heating element and the lower heating element being disposed in the interior chamber at opposing surfaces of the interior chamber can be determined based on the received temperature data, and the upper heating element and the lower heating element can be disposed in the interior chamber at opposing surfaces of the interior chamber. During an operating mode in which a third amount of heat is supplied to the interior chamber by the upper heating element and no heat is supplied to the interior chamber by the lower heating element, whether the temperature of the air within the upper region of the interior chamber exceeds a predetermined threshold can be determined. The first and second amounts of heat can be supplied to the interior chamber in response to determining that the temperature of the air within the upper region of the interior chamber exceeds the predetermined threshold.

(16) One or more of the following features can be included in any feasible combination. For example, mode data characterizing a mode of cooking the food can be received, and the first and second amounts of heat can be determined based on the received mode data. For example, the second and third amounts of heat can be adjusted after a period of time, and the period of time can be determined based on the temperature of the air within the upper region of the interior chamber and the temperature of the air within the lower region of the interior chamber. For example, the second and third amounts of heat can be adjusted when the temperature of the cooking surface is less than a target temperature. For example, the upper heating element and the lower heating element can be deactivated in response to determining that the temperature of the air within the upper region of the interior chamber exceeds the predetermined threshold. For example, a portion of the temperature data can be received from a lower temperature sensor disposed within the interior chamber and proximate the lower heating element and the cooking surface, and the lower temperature sensor can be configured to measure the temperature of at least one of the lower region of the interior chamber and the cooking surface.

(17) The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) These and other features will be more readily understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1A is a perspective view of a cooking device according to an embodiment;

(3) FIG. 1B is a first perspective cross-sectional view of the cooking device of FIG. 1A;

(4) FIG. 1C is a second perspective cross-sectional view of the cooking device of FIG. 1A;

(5) FIG. 1D is a third perspective cross-sectional view of the cooking device of FIG. 1A;

(6) FIG. 1E is a front side view of a user interface located on a front side of the cooking device of FIG. 1A;

(7) FIG. 1F is a schematic diagram illustrating components of the cooking device of FIG. 1A;

(8) FIG. 2 is a process flow diagram illustrating an exemplary process for initializing a pre-heat function of the cooking device of FIG. 1A;

(9) FIG. 3A is a process flow diagram illustrating an exemplary process for pre-heating the cooking device of FIG. 1A;

(10) FIG. 3B is a process flow diagram illustrating an exemplary process for determining a level of power to be applied to the heating elements of the cooking device of FIG. 1A; and

(11) FIG. 4 is a process flow diagram illustrating an exemplary process for cooking a pizza with the

cooking device of FIG. 1A.

(12) It is noted that the drawings are not necessarily to scale. The drawings are intended to depict only typical aspects of the subject matter disclosed herein, and therefore should not be considered as limiting the scope of the disclosure.

DETAILED DESCRIPTION

(13) Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings, and additional examples of specific system structures, functions, manufactures, uses, and related methods can be found in U.S. application Ser. Nos. 17/733,237, 17/663,582 and 18/079,781, each of which is incorporated by reference herein in its entirety. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.

(14) Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape.

(15) In general, cooking devices having at least one electric heating element and that are configured to cook various foods, including pizza, are provided. In some embodiments, the cooking device can include a housing having a base and a movable door coupled to the base that together define an interior cooking chamber. An upper heating element can be disposed within the interior chamber, and a lower heating element can be similarly disposed within the interior chamber opposite the upper heating element. A cooking surface, such as a cooking stone, can be disposed above the lower heating element within the cooking chamber such that food, such as a pizza, can be placed on top of the cooking stone when inserted into the interior chamber. The heating elements can be in operable communication with a controller configured to adjust the amount of heat supplied by the heating elements to the interior chamber to optimize the cooking of a food, such as pizza, placed inside the interior chamber.

(16) FIGS. 1A-1F illustrate an exemplary cooking device **10** having at least one electric heating element according to an embodiment. The cooking device **10** can be used to cook food in a variety of cooking modes, including conductive and convective modes (e.g., Warm, Dehydrate, Smoker, Bake, Roast, Broil, Hi/Low, Sear, Pizza, etc.). The illustrated cooking device **10** includes a base **12** and a movable door **14** movably coupled to the base **12**, such as via a hinge (not shown). Together, the walls of the base **12** (e.g., a rear wall **12R**, a left wall **12L**, a right wall **12RI**, a top wall **12T**, and a bottom wall **12B** as shown in FIGS. 1A-1D), and the door **14** can be referred to generally as a housing and can define an interior cooking chamber **16** that is sized to receive a variety of food products and/or food containers (e.g., a baking tray, a rack, a cooking stone, etc.). The cooking device **10** can include a lower heating element **18A** disposed above a lower surface **12L** of the base **12** and an upper heating element **18B** that is disposed below an upper surface (not shown) of the base **12**. As referenced above, the base **12** can include a cooking surface **20**, such as a cooking stone (also referred to as a pizza stone), upon which a food product (e.g., a pizza) can be placed during an operating procedure of the cooking device **10** and which can be disposed above the lower heating element **18A**. Each of the lower heating element **18A** and the upper heating element **18B** is configured to heat the cooking surface **20**, the interior cooking chamber **16** and/or the food product

through conduction, convection, radiation, or a combination thereof. In some embodiments, the lower heating element **18A** can comprise a plurality of lower heating elements, and, in some embodiments, the upper heating element **18B** can comprise a plurality of upper heating elements. The cooking device **10** can also include a fan **22**, which can be operated to circulate air within the interior cooking chamber **16** during a variety of cooking modes. The fan **22** can be located within the interior cooking chamber **16** at a rear wall of the base **12**, as shown in FIGS. **1C** and **1D**. The cooking device **10** can also include a lower temperature sensor **24A** disposed below the cooking surface **20** and configured to measure a temperature of the air in a lower region **16A** of the interior cooking chamber **16** that is proximate a lower surface of the cooking surface **20** (not shown). The lower surface of the cooking surface **20** is disposed opposite an upper surface **20A** of the cooking surface **20** as shown in FIG. **1C**, and the lower region **16A** can, in some embodiments, be bounded by the bottom wall **12B**, the left wall **12L**, the right wall **12RI**, the rear wall **12R**, the door **14**, and the lower surface of the cooking surface **20**. In some embodiments, the lower temperature sensor **24A** can be configured to measure a temperature of the lower surface of the cooking surface **20**. As such, in some embodiments, lower temperature data characterizing the lower surface of the cooking surface **20** and/or the air proximate the lower surface of the cooking **20** can be acquired and/or generated by the lower temperature sensor **24A** and provided to a controller (such as a controller **65**, described in further detail below) for use by the controller in determining the temperature of the air proximate the lower surface of the cooking surface **20** and/or the temperature of the lower surface of the cooking surface **20**. Similarly, the cooking device **10** can also include an upper temperature sensor **24B** disposed in the interior cooking chamber **16** and configured to measure a temperature of the air in an upper region **16B** of the interior cooking chamber **16** that is located above the cooking surface **20**. In some embodiments, the upper region **16B** can be bounded by the top wall **12T**, the left wall **12L**, the right wall **12RI**, the rear wall **12R**, the door **14**, and an upper surface **20B** of the cooking surface **20**. As such, in some embodiments, upper temperature data characterizing the air in the upper region **16B** of the interior cooking chamber **16** can be acquired and/or generated by the upper temperature sensor **24B** and provided to a controller (such as the controller **65** described below) for use by the controller in determining the temperature of the air proximate the lower surface of the cooking surface **20**. Exemplary lower and upper temperature sensors can include thermometers, thermocouples, NTC thermistors, and the like, or any collection/combination thereof. Lower and upper temperature sensors **24A**, **24B** are schematically illustrated in FIG. **1F**, and their use is described in further detail below.

(17) As explained above and as will be discussed in further detail below, the cooking device **10** can be pre-programmed to perform a unique cooking operation for cooking pizza based on a particular type or style of pizza, including Neapolitan style, thin-crust style, New York style, and pan style. An example of a Neapolitan style pizza can be a style of pizza featuring a relatively moist inner portion that includes pizza sauce (e.g., tomato-based sauce and the like) and pizza toppings (e.g., cheese, meat, vegetables and the like) and that is surrounded by a crust that has been subjected to a high amount of heat to thereby cause “leoparding” of the crust (e.g., the formation of relatively small blisters of heavily-cooked regions of the crust and/or larger-sized bubbles of heavily-cooked regions of the crust). In some circumstances, it can be difficult to satisfactorily cook a Neapolitan style pizza because of the difference between an amount of heat required to sufficiently cook the inner portion of the pizza and the high amount of heat required to cause the leoparding of the crust described above. For example, by failing to adequately control the high amount of heat required for leoparding, or by applying the high amount of heat required for leoparding to the crust to the inner portion, the inner portion can become overcooked and thus unsuitable for eating.

(18) The cooking device **10** can also be pre-programmed to perform a unique cooking operation for cooking a frozen pizza. In other aspects, the cooking device **10** can be configured to cook a pizza in accordance with user-defined custom cooking parameters. As such, in some embodiments, the cooking device **10** can include a user interface **50** that permits a user to configure the cooking

device **10** for performing one or more pre-programmed pizza cooking modes corresponding to one or more of the above-described styles/types of pizza. As explained above, in some embodiments, these pre-programmed pizza cooking modes can include “Neapolitan” (for cooking Neapolitan style pizza), “Thin” (for cooking thin-crust style pizza), “NY” (for cooking New York style pizza), “Pan” (for cooking pan style pizza), “Frozen” (for cooking frozen pizza), and “Custom” (for cooking a pizza in accordance with user-defined custom cooking parameters). Any of the listed operations can be used in combination with one another, both in succession or at the same time. Further, these cooking modes can be combined with smoke generated by a smoke generation unit **26** (illustrated in FIGS. **1A**, **1C**, and **1D**) integrated into the base **12**. The structure and/or functionality of an exemplary embodiment of such a smoke generation unit is described in greater detail in U.S. application Ser. Nos. 17/733,237, 17/663,582 and 18/079,781 (incorporated by reference herein as described above). The user interface **50** can be located on an external portion of the cooking device **10**, such as on a front face of the base **12**, as seen, for example, in FIGS. **1A-1D**. The user interface **50**, which is shown in detail in FIG. **1E**, can be in operable communication with a controller **65** (schematically illustrated in FIG. **1F**). The user interface can be configured to receive inputs that cause the controller **65** to perform one or more actions responsive to the received inputs, as described in further detail below.

(19) In some embodiments, the user interface **50** can include a variety of input buttons and controls that permit the user to configure the operation of the cooking device **10** by providing the above-described inputs. For example, in some embodiments, the user interface **50** can include a rotary function dial **51** which is configured to rotate when turned by a user, such that the user can select a pre-programmed cooking mode in which the cooking device **10** can operate. Exemplary pre-programmed cooking modes can include Warm, Dehydrate, Smoker, Bake, Roast, Broil, Hi/Low, Sear, and Pizza. The user interface **50** can also include a smoke generation button **52a** which can cause the controller **65** to activate the smoke generation unit **26** described above and thereby supply smoke to the cooking chamber **16**. The user interface **50** can illuminate a smoke generation icon **52b** when the smoke generation unit **26** is activated. The user interface **50** can also include temperature and time buttons **53**, **54**, which, when pressed, can allow the user to adjust the cooking temperature and the cooking time of the cooking device **10**, which are displayed above the buttons **53**, **54** as illustrated in FIG. **1E**. The user interface **50** can also include a start/stop button **55**, which, when pressed by a user, causes the controller **65** to start and/or stop various operations of the cooking device **10** (e.g., pre-heating, cooking, etc.). The user interface **50** can also include a start/stop dial **56** which allows the user to perform additional functionality, such as scrolling through the pizza cooking modes referenced above and as described in further detail below. The user interface **50** can also include one or more indicators which indicate a status of the cooking device **10**. For example, the user interface **50** can include a pizza cooking mode indicator **57**, which can illuminate an indicator corresponding to one of the above-described pizza cooking modes when that mode is selected. Similarly, the user interface **50** can also include an “On” indicator **58** which illuminates when the cooking device is in an operational state.

(20) As schematically illustrated in FIG. **1F** and as referenced above, the cooking device **10** can include a controller **65** that is in operable communication with one or more of the components described above (e.g., the user interface **50**, the lower temperature sensor **24A**, the upper temperature sensor **24B**, etc.). As shown, the controller **65** can include at least one processor **66** and a memory **67** storing instructions which, when executed by the processor **66**, can cause the at least one data processor to perform one or more of the operations described elsewhere herein. The controller **65** can also include an input/output (I/O) interface **68** that enables the processor **66** to receive commands and/or data from other components of the cooking device **10** for use in performing the operations. For example, the controller **65** can receive, through the I/O interface **68**, data characterizing temperature measurements made by the lower temperature sensor **24A** and/or the upper temperature sensor **24B** and provide that data to the processor **66** for use in performing

operations requiring the received data as an input. Similarly, the controller **65** can receive, from the user interface **50** and via I/O interface **68**, data characterizing inputs received from the user by the user interface **50** and provide that data to the processor **66** for use in performing operations that require the data received from the user interface **50** as input.

(21) As shown in FIG. **1F**, the cooking device **10** can also include a power supply **70** that is configured to supply power to the lower heating element **18A** and the upper heating element **18B** (in addition to other components of the cooking device **10** requiring power to operate). The power supply **70** can be in operable communication with the controller **65**. As such, the power supply **70** can be configured to receive commands from the controller **65** (provided via the I/O interface **68**) that cause the power supply **70** to provide electrical power to the lower heating element **18A** and/or the upper heating element **18B** to thereby cause the lower heating element **18A** and/or the upper heating element **18B** to activate and provide heat to the cooking chamber **16**. In some embodiments, the power delivered to the lower heating element **18A** and/or the upper heating element **18B** can vary based on the commands received from the controller **65**. For example, the level and/or duration of power supplied to the lower heating element **18A** and/or the upper heating element **18B** can vary over a given period of time based on power delivery instructions characterized by the commands received from the controller **65**. As a result, the level and/or duration of heat being supplied to the cooking chamber **16** can vary over the given period of time based on the commands received from the controller **65**.

(22) In some cooking scenarios, it can be advantageous to “pre-heat” the cooking device **10** prior to inserting the pizza into the interior cooking chamber **16**. FIG. **2** illustrates an exemplary process **200** for the initializing of a pre-heat function of the cooking device **10** and which can be implemented in some embodiments of the present subject matter. As shown, the process **200** begins at **201** when a user has selected the “Pizza” cooking mode with the function dial **51**. At **201**, the smoke generation icon **52b** on the user interface **50** is not illuminated (which indicates that the smoke generation unit **26** is not operational), and the user interface **50** displays the default cooking time. The process then proceeds to **202**, at which point the controller **65** monitors the position of the function dial **51** and determines whether the user is rotating the function dial **51**. In response to determining that the user is rotating the function dial **51**, the controller **65** then enters a function adjustment mode **203** in which the controller configures one or more components of the cooking device **10** in accordance with a selected one of the above-described cooking modes.

(23) In response to determining that the user is not rotating the function dial **51**, the controller monitors the position of the start/stop dial **56** and determines, at **204**, whether the user is rotating the start/stop dial **56**. If the controller determines that the user is rotating the start/stop dial **58**, the controller then enters a pizza function adjustment mode **205**. When the controller enters the pizza function adjustment mode **205**, all available pizza cooking modes are illuminated on the user interface **50**. By rotating the start/stop dial **58**, the user can scroll through each of the pizza cooking modes (e.g., Neapolitan, Thin, NY, Pan, Frozen, and Custom) and thereby change the selected pizza cooking mode in accordance with their preference. In some embodiments, the controller **65** can also cause the user interface **50** to modify one or more aspects of the display based on the selected pizza cooking mode. For example, as the user rotates the start/stop dial **58** and scrolls through each of the pizza cooking modes, the controller **65** can update the time and/or temperature displayed on the user interface **50** to reflect a default length of cooking time and a default cooking temperature, respectively, that corresponds to the pizza cooking mode currently selected.

(24) In response to determining, at **204**, that the user is not rotating the start/stop dial **58**, the controller **65** determines, at **206**, whether the user has pressed the time button **53**. If the controller determines that the user has pressed the time button **53**, the controller then enters a time adjustment mode **207** in which the user can interact with the user interface **50** to adjust the length of the cooking time from the default length of the cooking time associated with the selected pizza cooking mode to a desired length of the cooking time.

(25) In response to determining, at **206**, that the user has not pressed the time button, the controller **65** determines, at **208**, whether the user has selected the custom function by rotating the start/stop dial **56** as discussed above. If the controller **65** determines that the user has selected the custom function, the controller then enters a custom adjustment mode **209**. When the controller is in the custom adjustment mode **209**, the user can adjust the cooking temperature and the length of the cooking time, and the user can set a single temperature value that serves as a set point for the temperature of the cooking chamber as well as a set point for the temperature of the bottom of the cooking surface **20**. When the user adjusts the cooking temperature such that the set point for the temperature of the bottom of the cooking surface **20** exceeds a first predetermined temperature threshold, the controller can determine an optimized bottom set point temperature based on the cooking temperature selected by the user.

(26) In response to determining, at **208**, that the user has not selected the custom function, the controller **65** determines, at **210**, whether the user has pressed a probe button located on the user interface **50** (not shown). If the controller **65** determines that the user has pressed a probe button, the controller, at **211**, causes an error notification to be presented to the user.

(27) In response to determining, at **210**, that the user has not pressed the probe button, the controller **65** determines, at **212**, whether the user has pressed the smoke generation button **52a**. If the controller **65** determines that the user has pressed the smoke generation button **52a**, the controller then causes a smoke operating mode to activate **213**. The controller **65** can determine, at **214**, whether the user has pressed the smoke generation button **52a** a second time and cause, at **215**, the smoke generation unit **26** to exit the smoke operating mode in response to the user pressing the smoke generation button **52a** a second time.

(28) After making the determinations described above at **212** and **214**, the controller **65** can determine, at **216**, whether the user has pressed the start/stop button **55**. If the controller **65** determines that the user has not pressed the start/stop button **55**, the process then returns to **202**, and the controller **65** monitors the position of the function dial **51** to determine whether the user has rotated the function dial **51** as described above.

(29) If the controller **65** determines that the user has pressed the start/stop button **55**, the controller **65** determines, at **217** whether the configuration of the cooking device **10** (e.g, the previously-determined settings/functions) is the same as that which was active during a previous period of time (e.g., the immediately preceding five minutes, etc.). If so, the pre-heat function is skipped and the initialization process **200** ends at **218**.

(30) If the controller **65** determines that the configuration of the cooking device **10** is not the same as that which was active during the previous period of time, the controller **65** determines at **219** whether the smoke mode has been activated. If so, the controller **65** causes the smoke generation unit **26** to initiate a fuel ignition process **220**, an example of which is described in U.S. patent application Ser. No. 17/663,582, hereby incorporated by reference herein in its entirety, and, once complete, enables the cooking device **10** for pre-heating as described below with respect to FIG. 3A, and the initialization process ends at **222**. If not, the controller **65** enables the cooking device **10** for pre-heating as described below with respect to FIG. 3A and the initialization process ends at **223**. In some embodiments, if the fuel ignition process **220** has previously occurred, the pre-heating function is skipped.

(31) FIG. 3A illustrates an exemplary process **300A** for controlling the pre-heating of the cooking device **10** and which can be implemented in some embodiments of the present subject matter. The process **300** begins at **301**, wherein the initialization process **200** as described above with respect to FIG. 2 has ended at either of **222** or **223**.

(32) At **301**, the controller **65** has begun to pre-heat the cooking chamber **16** and has updated the user interface **50** to indicate that the cooking device **10** is pre-heating the cooking chamber **16** in accordance with the pizza function selected by the user as described above. At **302**, the controller **65** determines the amount of power that the power supply **70** should provide the lower heating

element **18A** and/or the upper heating element **18B** and subsequently provides instructions to the power supply **70** that cause the power supply **70** to provide the determined amounts of power to be supplied to each of the lower and upper heating elements **18A**, **18B** (and thereby cause the pre-heating of the cooking chamber **16**). The process by which the controller **65** determines the amount of power that the power supply **70** should provide to the lower heating element **18A** and to the upper heating element **18B** is described in detail below with respect to FIG. 3B.

(33) While the cooking device is undergoing pre-heating, the controller **65** determines, at **303**, whether the pre-heating of the cooking chamber **16** has occurred for a first predetermined amount of time (e.g., 35 minutes, etc.). In response to determining that the pre-heating of the cooking chamber **16** has occurred for the first predetermined amount of time, the controller **65** enables the cooking device **10** for the cooking of pizza as described below with respect to FIG. 4, and the pre-heating process **300** ends at **304**. In response to determining that the pre-heating of the cooking chamber **16** has not occurred for the first predetermined amount of time, the controller **65** determines, at **305**, whether the pre-heating of the cooking chamber **16** has occurred for a second predetermined amount of time (e.g., 25 minutes, etc.) that is less than the first predetermined amount of time. If the controller **65** determines that the pre-heating of the cooking chamber **16** has occurred for the second predetermined amount of time, the controller **65** determines, at **306**, whether the cooking temperature is less than or equal to a second predetermined temperature threshold. In some embodiments, the second predetermined temperature threshold may be the same as the first determined temperature threshold described above. If the controller **65** determines, as part of the power amount determinations performed at **302**, that the pre-heating process is complete, the controller **65** can enable the cooking device **10** for the cooking of pizzas as described below with respect to FIG. 4 and end the pre-heating process at **307**.

(34) As explained above, at **302** of the pre-heating control process **300A**, the controller **65** can determine the amount of power that should be provided to each of the lower heating element **18A** and the upper heating element **18B** by the power supply **70**. FIG. 3B illustrates an exemplary power output calculation process **300B** to be performed at **302** and which can be implemented in some embodiments of the present subject matter. As shown, at the beginning of the process **302**, the controller **65** determines, at **308**, whether the temperature measured with the upper temperature sensor **24B** is greater than or equal to a target temperature. If the controller **65** determines that the temperature measured with the upper temperature sensor **24B** is greater than or equal to the target temperature, the controller **65** then determines, at **309**, whether the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature. If the controller **65** determines that temperature measured with the lower temperature sensor **24A** is less than the target temperature, the controller **65** then, at **310**, causes the power supply **70** to provide no power to the upper heating element **18B** and to provide full power to be supplied to the lower heating element **18A**. If the controller **65** determines, at **308**, that the temperature measured with the upper temperature sensor **24B** is less than the target temperature, the controller then determines, at **311**, whether the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature. If the controller **65** determines at **311** that the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature, the controller **65** then, at **312**, causes the power supply **70** to provide no power to the lower heating element **18A** and full power to be supplied to the upper heating element **18B**.

(35) If the controller **65** determines at **311** that the temperature measured with the lower temperature sensor **24A** is less than the target temperature, the controller **65** then, at **313**, determines a power ratio characterizing the relationship between the amount of power to be supplied to the upper heating element **18B** relative to the amount of power to be supplied to the lower heating element **18A**. In some embodiments, the power ratio can characterize a discrepancy between an actual temperature, measured by a temperature sensor (such as the lower temperature sensor **24A** and/or the upper temperature sensor **24B**), and a target temperature. To determine the

power ratio, in some embodiments and as explained above, the controller **65** can receive, from the upper temperature sensor **24B**, upper temperature data characterizing the temperature of the air within the cooking chamber **16**, and the controller **65** can receive, from the lower temperature sensor **24A**, lower temperature data characterizing the temperature of the bottom of the cooking surface **20**. The controller **65** can determine the power ratio based on the received upper temperature data, the received lower temperature data, and the target temperature.

(36) The controller **65** also, at **313**, determines a lower heating element energy value, which characterizes a portion of available energy that is dedicated to the lower heating element **18A** and that is above and beyond a predetermined minimum lower heating element power value when the temperature measured with the lower temperature sensor **24A** and the temperature measured with the upper temperature sensor **24B** are below their respective target temperatures. Additionally, the controller **65**, at **313**, determines an upper heating element energy value, which characterizes a portion of available energy that is dedicated to the upper heating element **18B** and that is above and beyond a predetermined minimum upper heating element power value when the temperature measured with the lower temperature sensor **24A** and the temperature measured with the upper temperature sensor **24B** are below their respective target temperatures. The controller **65** determines the lower heating element energy value based on the determined power ratio, the minimum lower heating element power value, and the minimum upper heating element power value. Additionally, the controller **65** determines the upper heating element energy value based on the minimum lower heating element power value, the minimum upper heating element power value, and the determined lower heating element energy value.

(37) The controller **65**, at **313**, determines a lower heating portion value that characterizes a portion of the available energy allocated to the lower heating element **18A** when each of the temperature measured with the lower temperature sensor **24A** and the temperature measured with the upper temperature sensor **24B** are below their respective target temperatures. And, similarly, the controller **65**, at **313**, determines an upper heating portion value that characterizes a portion of the available energy allocated to the upper heating element **18B** when each of the temperature measured with the lower temperature sensor **24A** and the temperature measured with the upper temperature sensor **24B** are below their respective target temperature. The controller **65** determines the lower heating portion value based on the minimum lower heating element power value and the determined lower heating element energy value, and the controller **65** can determine the upper heating portion value based on the minimum upper heating element power value and the determined upper heating element energy value.

(38) In some embodiments, the controller **65** can determine the lower heating element energy value without regard to the predetermined minimum lower heating element power value. In this case, the lower heating element energy value characterizes a portion of available energy that is dedicated to the lower heating element **18A**. As such, the controller **65** can determine the lower heating element energy value based on the determined power ratio. And, as such, the controller **65** can determine the lower heating portion value based on the determined lower heating element energy value. Similarly, in some embodiments, the controller **65** can, at **313**, determine the upper heating element energy value without regard to the predetermined minimum upper heating element power value. In this case, the upper heating element energy value characterizes a portion of available energy that is dedicated to the upper heating element **18B**. As such, the controller **65** can determine the upper heating element energy value based on the determined lower heating element energy value. And, as such, the controller **65** can determine the upper heating portion value based on the determined upper heating element energy value.

(39) Once the controller **65** has made the above-described determinations at **313**, the controller **65**, at **314**, causes the lower heating element **18A** to operate at full power and the upper heating element **18B** to be deactivated. At **315**, the controller **65** determines whether the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature.

If the controller **65** determines that the temperature measured with the lower temperature sensor **24A** is less than the target temperature, the controller **65** then determines, at **316** and based on the lower heating portion value, a portion of a predefined time interval during which the lower heating element **18A** is activated and whether that portion of the predefined time interval has elapsed. If so, the controller **65**, at **317**, causes the lower heating element **18A** to deactivate and the upper heating element **18B** to activate at full power. If not, the controller **65** re-determines, at **315**, whether the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature. If, at **315**, the controller **65** determines that the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature, the controller bypass the determination at **316** and causes the lower heating element **18A** to deactivate and the upper heating element **18B** to activate at full power at **317**.

(40) At **318**, the controller **65** receives data characterizing temperatures measured by the upper temperature sensor **24B** and determines, based on the received temperature data, whether the temperature of the air measured with the upper temperature sensor **24B** is greater than or equal to the target temperature. If the controller **65** determines at **318** that the temperature measured with the upper temperature sensor **24B** is less than the target temperature, the controller **65** then determines, at **319** and based on the upper heating portion value, a portion of a predefined time interval during which the upper heating element **18B** is activated and whether that portion of the predefined time interval has elapsed. If so, the controller **65** then returns to **308** to re-determine whether the temperature measured with the upper temperature sensor **24B** is greater than or equal to the target temperature and performs one or more of the operations based on the results of that determination as described elsewhere herein. If not, the controller **65** then returns to **318** to re-determine whether the temperature measured with the upper temperature sensor **24B** is greater than or equal to the target temperature. If the controller **65** determines, at **318**, that the temperature measured with the upper temperature sensor **24B** is greater than or equal to the target temperature, the controller **65** deactivates the upper heating element **18B** and the lower heating element **18A** at **320**. At **321**, the controller **65** determines whether the cooking device **10** is operating in a pre-heat state. If so, the system ends the pre-heating process at **304** (as also illustrated in FIG. 3A). If not, the controller **65**, at **322**, updates the user interface **50** to display the default temperature setting for the selected pizza function as well as a countdown timer that indicates the amount of time remaining during the pre-heating process.

(41) As explained above, the controller **65**, at **309**, determines whether the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature. If the controller **65** determines that the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature, the controller **65** then performs the functionality described above at **321**.

(42) Additionally, as explained above, if, at **309**, the controller **65** determines that the temperature measured with the lower temperature sensor **24A** is less than the target temperature, the controller **65** then, at **310**, causes no power to be supplied to the upper heating element **18B** and full power to be supplied to the lower heating element **18A**. When this occurs, the controller **65** determines, at **323**, whether the lower heating element **18A** has been activated for a first predefined period of time (e.g., 30 seconds). If the controller **65** determines that the lower heating element **18A** has been activated for the first predefined period of time, the controller **65** returns to **308** to re-determine whether the temperature the temperature measured with the upper temperature sensor **24B** is greater than or equal to the target temperature. If the controller **65** determines that the lower heating element **18A** has been activated for less than the first predefined period of time, the controller **65** then, at **324**, determines whether the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature. If so, the controller **65**, at **325**, causes the lower and upper heating elements **18A**, **18B** to deactivate and then performs the functionality described above at **321**.

(43) In addition, as explained above, if the controller **65** determines, at **311**, that the temperature measured with the lower temperature sensor **24A** is greater than or equal to the target temperature, the controller **65** then, at **312**, causes no power to be supplied to the lower heating element **18A** and full power to be supplied to the upper heating element **18B**. When this occurs, the controller **65** determines, at **326**, whether the upper heating element **18B** has been activated for a second predefined period of time (e.g., 30 seconds). If the controller **65** determines that the upper heating element **18B** has been activated for a second predefined period of time, the controller **65** returns to **308** to re-determine whether the temperature the temperature measured with the upper temperature sensor **24B** is greater than or equal to the target temperature and performs one or more of the operations based on the results of that determination as described elsewhere herein. If the controller **65** determines that the lower heating element **18A** has been activated for less than the second predefined period of time, the controller **65** then, at **327**, determines whether the temperature measured with the upper temperature sensor **24B** is greater than or equal to the target temperature. If so, the controller **65**, at **328**, turns off the upper and lower heating elements **18A**, **18B** and then performs the functionality described above at **321**.

(44) However, in some embodiments, the pre-heat function can be skipped in response to the occurrence of one or more events. For example, in some embodiments, the pre-heat function can be skipped by a user pushing and holding down the start/stop dial **56** for a certain period of time. And, as another example, in some embodiments, the pre-heat function can be skipped if the temperature of the air within the cooking chamber exceeds a target temperature for a particular pizza cooking mode at the time the pizza cooking mode is selected as described above.

(45) FIG. **4** illustrates an exemplary process **400** for controlling the pizza cooking function of the cooking device **10** and which can be implemented in some embodiments of the present subject matter. In some embodiments, process **400** can occur after the pre-heating process **300a** described above has concluded. The process **400** begins at **401**, at which point the controller **65** causes power to be supplied as appropriate to the upper heating element **18B** and the lower heating element **18A** and such that the pre-heat set temperature is maintained in accordance with the pizza cooking mode selected in the manner described above. The controller **65** then, at **402**, determines whether the user has pressed the start/stop button **55** on the user interface **50**. If not, the controller **65** then, at **403**, determines whether a certain period of time (e.g., an hour, etc.) has elapsed since the beginning of process **400** without the user having pressed the start/stop button **55**. If not, the user interface displays a periodic notification to the user. If so, the controller **65** causes the cooking device **10** to deactivate, ending the process at **404**.

(46) In response to determining at **402** that the user has pressed the start/stop button **55**, then, the controller **65** then activates one or more of the heating elements to begin to cook the pizza at **405**. The controller **65** causes the upper and/or lower heating elements **18A**, **18B** to activate in accordance with the pizza function selected by the user. For example, if the user has selected Neapolitan as the pizza cooking mode, the controller **65** only activates the upper heating element **18B** and deactivates the lower heating element **18A**. For example, if the user has selected one of the other pizza cooking modes described above, the controller **65** determines the amount of power that the power supply **70** is to provide to the upper heating element **18B** and to the lower heating element **18A** independently by executing the process **300B** described above with respect to FIG. **3B**. Once the controller **65** has made that determination, the controller **65** then causes the power supply **70** to supply the determined amount of power to each of the upper heating element and the lower heating element **18A**.

(47) If the user has selected the Neapolitan pizza function, the controller **65** determines, at **406**, whether the temperature measured with the upper temperature sensor **24B** exceeds a first predetermined threshold (e.g., 750 degrees Fahrenheit). If so, then the controller **65** causes the upper heating element **18B** to deactivate and the lower heating element **18A** to activate at full power at **407**. After adjusting the operation of the heating elements at **407** and when in the

Neapolitan mode, the controller **65** determines, at **408**, whether the temperature measured with the upper temperature sensor **24B**, as measured by the upper temperature sensor **24B**, exceeds a second predetermined threshold (e.g., 700 degrees Fahrenheit). If not, then, the controller **65** determines, at **409**, whether the temperature measured with the lower temperature sensor **24A** exceeds a predetermined threshold (e.g., 700 degrees Fahrenheit). If the controller **65** determines that the temperature measured with the lower temperature sensor **24A** has exceeded the predetermined threshold, (e.g., 700 degrees Fahrenheit), then the controller **65**, at **410**, deactivates the lower heating element **18A** and the upper heating element **18B** (if not previously deactivated), determines whether a predetermined period of time (e.g., 5 seconds) has passed, and returns to **408** to re-determine whether the temperature of the air in the cooking chamber **16** exceeds the first predetermined threshold (e.g., 700 degrees Fahrenheit). If not, the controller returns to **408** to monitor the temperature of the air in the cooking chamber **16** and re-determine whether the temperature measured with the upper temperature sensor **24B** exceeds the first predetermined threshold (e.g., 700 degrees Fahrenheit). In some embodiments, the controller can reactivate the lower heating element **18A** and/or the lower heating element **18B** in response to determining whether the temperature measured with the lower temperature sensor **24A** and/or the temperature measured with the upper temperature sensor **24B** has exceeded a predetermined threshold. In some embodiments, if the controller **65** determines that the temperature measured with the upper temperature sensor **24B** has exceeded the predetermined threshold, (e.g., 700 degrees Fahrenheit), then the controller **65** deactivates the lower heating element **18A** and the upper heating element **18B** (if not previously deactivated) and re-executes the process **300B** described above to pre-heat the cooking chamber **16** as quickly as possible.

(48) During the period of time that the cooking device **10** is cooking the pizza, a user can press the start/stop button **55** on the user interface **50** to pause cooking. As such, during cooking, if the controller **65** determines, at **412**, that the start/stop button **55** has been pressed by the user, the controller **65** causes, at **413**, the heating elements **18A** and/or **18B** to deactivate and a notification to be displayed on the user interface **50** indicating that the cooking process has ended and instructing the user to add additional food to the cooking chamber **16** and to resume cooking by pressing the start/stop button **55**. When this occurs, the controller **65**, at **414**, re-executes the process **300B** described above to maintain the level of heat required for the selected pizza cooking mode in the cooking chamber **16**.

(49) If the controller **65** determines at **412** that the user has not pressed the stop/start button **55** when the pizza is being cooked, cooking continues in accordance with the time and temperature settings of the selected pizza cooking mode. As the cooking device **10** continues to cook the pizza, the controller **65** monitors the length of time during which the pizza is being cooked. When the controller **65** determines, at **415**, that the time allotted for cooking of the pizza and specified by the selected pizza cooking mode has elapsed, the controller **65**, at **416**, causes a notification to be displayed on the user interface indicating that the pizza has been cooked and instructing the user to remove the cooked pizza. Once this occurs, the controller **65**, at **414**, re-executes the process **300B** described above to pre-heat the cooking chamber **16** as quickly as possible following the cooking of the first pizza. After initializing the second pre-heat process at **414**, the controller **65** monitors the temperature measured with the lower temperature sensor **24A** and the temperature measured with the upper temperature sensor **24B** and determines whether those temperatures have reached their respective target temperatures at **417**. In response to determining, at **417**, that both target temperatures have been met, the controller **65** causes a notification to appear on the user interface **50** indicating that the second pre-heat process has ended at **419** and that the cooking device **10** is ready to end the pre-heating process described above and cause the cooking process **400** to begin once again at **401** in response to determining that a user has the start/stop button **55** on the user interface **50** at **420**. If the controller **65** determines, at **420**, that a user has failed to press the start/stop button to restart the cooking process, and, at **421**, that a predefined period of time (e.g.,

an hour) has elapsed since the end of the second pre-heating process, and, at 22, that no other cooking process is actively running, the controller 65 causes the cooking device to turn off at 423.

(50) One or more aspects or features of the subject matter described herein can be realized in digital electronic circuitry, integrated circuitry, specially designed application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs) computer hardware, firmware, software, and/or combinations thereof. These various aspects or features can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which can be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device. The programmable system or computing system may include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

(51) These computer programs, which can also be referred to as programs, software, software applications, applications, components, or code, include machine instructions for a programmable processor, and can be implemented in a high-level procedural language, an object-oriented programming language, a functional programming language, a logical programming language, and/or in assembly/machine language. As used herein, the term “machine-readable medium” refers to any computer program product, apparatus and/or device, such as for example magnetic discs, optical disks, memory, and Programmable Logic Devices (PLDs), used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor. The machine-readable medium can store such machine instructions non-transitorily, such as for example as would a non-transient solid-state memory or a magnetic hard drive or any equivalent storage medium. The machine-readable medium can alternatively or additionally store such machine instructions in a transient manner, such as for example as would a processor cache or other random access memory associated with one or more physical processor cores.

(52) To provide for interaction with a user, one or more aspects or features of the subject matter described herein can be implemented on a computer having a display device, such as for example a cathode ray tube (CRT) or a liquid crystal display (LCD) or a light emitting diode (LED) monitor for displaying information to the user and a keyboard and a pointing device, such as for example a mouse or a trackball, by which the user may provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well. For example, feedback provided to the user can be any form of sensory feedback, such as for example visual feedback, auditory feedback, or tactile feedback; and input from the user may be received in any form, including, but not limited to, acoustic, speech, or tactile input. Other possible input devices include, but are not limited to, touch screens or other touch-sensitive devices such as single or multi-point resistive or capacitive trackpads, voice recognition hardware and software, optical scanners, optical pointers, digital image capture devices and associated interpretation software, and the like.

(53) Certain exemplary implementations have been described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the systems, devices, and methods disclosed herein. One or more examples of these implementations have been illustrated in the accompanying drawings. Those skilled in the art will understand that the systems, devices, and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary implementations and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary implementation may be combined with the features of other implementations. Such modifications and variations are intended to be included within the scope of the present invention. Further, in the present disclosure,

like-named components of the implementations generally have similar features, and thus within a particular implementation each feature of each like-named component is not necessarily fully elaborated upon.

(54) Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

(55) One skilled in the art will appreciate further features and advantages of the invention based on the above-described implementations. Accordingly, the present application is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated by reference in their entirety.

Claims

1. A method comprising: receiving first temperature data characterizing a temperature of air within an upper region of an interior chamber of a cooking device, the upper region being above a cooking surface, and second temperature data characterizing a temperature of air within a lower region of the interior chamber, the lower region being below the cooking surface; determining, based on the received first and second temperature data, a ratio between an amount of heat to be supplied to the interior chamber by an upper heater and an amount of heat to be supplied to the interior chamber by a lower heater, the upper and lower heaters being disposed in the interior chamber at opposing surfaces of the interior chamber; causing heat to be supplied to the interior chamber by at least one of the upper and lower heaters based on the determined ratio; and adjusting an amount of heat supplied by at least one of the upper and lower heaters after a period of time, wherein the period of time is determined based on the determined ratio.
2. The method of claim 1, further comprising receiving mode data characterizing a mode of cooking food, wherein the mode data includes a target temperature, and wherein the ratio is further determined based on the received mode data.
3. The method of claim 1, the causing of the heat to be supplied to the interior chamber occurs when the cooking device is operating in a pre-heating state.
4. The method of claim 3, further comprising adjusting the amount of the heat supplied to the interior chamber by the upper heater and the amount of the heat supplied to the interior chamber by the lower heater when the temperature of the air within the lower region of the interior chamber is less than a target temperature.
5. The method of claim 4, further comprising: receiving third temperature data characterizing a subsequent temperature of the air within the upper region of the interior chamber; determining whether the subsequent temperature is greater than or equal to the target temperature; and in response to determining that the subsequent temperature is greater than or equal to the target temperature, causing the upper heater and the lower heater to deactivate.
6. The method of claim 2, wherein the method is carried out by a cooking system having a user interface, the method further comprising receiving, via the user interface, a user input characterizing a selection of the mode of cooking the food from a plurality of cooking modes.
7. The method of claim 6, wherein the plurality of cooking modes comprise one or more predefined modes for cooking pizza and a custom mode configured for cooking a pizza in accordance with user-defined custom cooking parameters.
8. The method of claim 7, wherein the period of time is further determined based on the mode data.

9. The method of claim 8, further comprising: comparing the temperature of air within the lower region to the target temperature; determining whether the period of time has elapsed; responsive to determining that the temperature of the air within the lower region is less than the target temperature and that the period of time has elapsed, causing the lower heater to deactivate and the upper heater to activate at a maximum power; and responsive to determining that the temperature of the air within the lower region is greater than or equal to the target temperature, causing the lower heater to deactivate and the upper heater to activate at the maximum power.

10. The method of claim 1, further comprising: determining a lower heater energy value based on the determined ratio, the lower heater energy value characterizing a portion of available energy allocated to the lower heater and which is higher than a predetermined minimum lower heater power value; determining a lower heating portion value based on the lower heater energy value, the lower heating portion value characterizing a portion of available energy allocated to the lower heater when a first temperature measured with a lower temperature sensor is below a first predetermined threshold temperature and a second temperature measured with an upper temperature sensor is below a second predetermined threshold temperature; and wherein the period of time is determined based on the determined lower heating portion value.

11. A cooking device comprising: an interior chamber having a cooking surface; a first temperature sensor configured acquire first temperature data characterizing a temperature of air within an upper region of the interior chamber, the upper region being above the cooking surface; a second temperature sensor configured acquire second temperature data characterizing a temperature of air within a lower region of the interior chamber, the lower region being below the cooking surface; a first heater configured to provide heat to the upper region; a second heater configured to provide heat to the lower region; and a controller communicatively coupled to the first and second temperature sensors and the first and second heaters, the controller configured to perform operations comprising receiving first temperature data characterizing a temperature of air within the upper region from the first temperature sensor and second temperature data characterizing a temperature of air within the lower region from the second temperature sensor, determining, based on the received first and second temperature data, a ratio between an amount of heat to be supplied to the interior chamber, causing heat to be supplied to the interior chamber by at least one of the first and second heaters based on the determined ratio, and adjusting an amount of heat supplied by at least one of the first and second heaters after a period of time, wherein the period of time is determined based on the determined ratio.

12. The cooking device of claim 11, wherein the controller is configured to perform operations further comprising receiving mode data characterizing a mode of cooking food, wherein the mode data includes a target temperature, and wherein the ratio is further determined based on the received mode data.

13. The cooking device of claim 11, wherein the causing of the heat to be supplied to the interior chamber occurs when the cooking device is operating in a pre-heating state.

14. The cooking device of claim 13, wherein the controller is configured to perform operations further comprising adjusting the amount of the heat supplied to the interior chamber by the first heater and the amount of the heat supplied to the interior chamber by the second heater when the temperature of the air within the lower region of the interior chamber is less than a target temperature.

15. The cooking device of claim 14, wherein the controller is configured to perform operations further comprising: receiving third temperature data characterizing a subsequent temperature of the air within the upper region of the interior chamber from the first temperature sensor; determining whether the subsequent temperature is greater than or equal to the target temperature; and in response to determining that the subsequent temperature is greater than or equal to the target temperature, causing the first heater and the second heater to deactivate.

16. The cooking device of claim 11, further comprising a user interface, wherein the controller is

configured to receive, via the user interface, a user input characterizing a selection of the mode of cooking the food from a plurality of cooking modes.

17. The cooking device of claim 16, wherein the plurality of cooking modes comprise one or more predefined modes for cooking pizza and a custom mode configured for cooking a pizza in accordance with user-defined custom cooking parameters.

18. The cooking device of claim 17, wherein the period of time is further determined based on the mode data.

19. The cooking device of claim 18, wherein the controller is configured to perform operations further comprising: comparing the temperature of air within the lower region to the target temperature; determining whether the period of time has elapsed; responsive to determining that the temperature of the air within the lower region is less than the target temperature and that the period of time has elapsed, causing the lower heater to deactivate and the upper heater to activate at a maximum power; and responsive to determining that the temperature of the air within the lower region is greater than or equal to the target temperature, causing the lower heater to deactivate and the upper heater to activate at maximum power.
